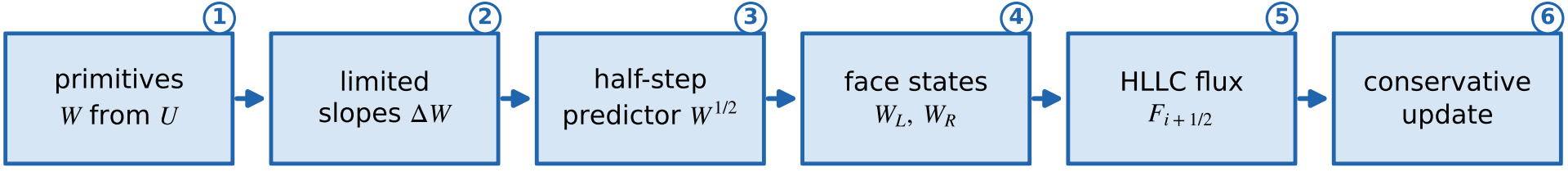
MUSCL-Hancock step (2nd order, unsplit)



1 $W = (\rho, \mathbf{v}, P), \quad P = (\gamma - 1)\left(E - \frac{1}{2}\rho v^2\right)$

2 $\Delta W_i = \text{minmod}\,(W_i - W_{i-1},\ W_{i+1} - W_i)$ per axis

3 $W_i^{1/2} = W_i - \frac{\Delta t}{2}\left(A_x \frac{\Delta_x W}{\Delta x} + A_y \frac{\Delta_y W}{\Delta y} + A_z \frac{\Delta_z W}{\Delta z}\right), \quad A = \partial F/\partial W$

4 $W_{L,\,i+1/2} = W_i^{1/2} + \frac{1}{2}\Delta W_i, \qquad W_{R,\,i+1/2} = W_{i+1}^{1/2} - \frac{1}{2}\Delta W_{i+1}$

5 $S_L = \min(u_L - c_L,\ u_R - c_R), \quad S_R = \max(\ldots), \quad S_* = \frac{P_R - P_L + \rho_L u_L (S_L - u_L) - \rho_R u_R (S_R - u_R)}{\rho_L (S_L - u_L) - \rho_R (S_R - u_R)}$

5 $F = F_L$ if $S_L > 0$; $F_L + S_L(U_L^* - U_L)$ if $S_* \geq 0$; mirrored right states otherwise

6 $U_i^{n+1} = U_i^n - \Delta t \sum_d \frac{1}{\Delta x_d}\left(F_{i+1/2}^{(d)} - F_{i-1/2}^{(d)}\right)$

unsplit (one time level) · ghosts carry BC / shard states · slope-limited ⇒ TVD, shock-capturing